HR's Notes
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SESSION-2 HTML NOTES

Unordered Lists

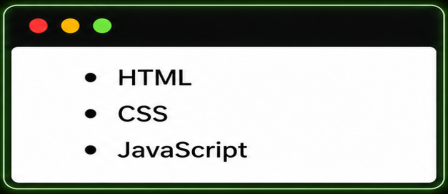
WHAT & WHY

Use `<ul>` when you have a collection of items where order does not matter semantically.
The browser renders bullet markers automatically. Swapping items doesn't change meaning.

e.g:

```
<ul>
   <li>HTML</li>
   <li>CSS</li>
   <li>JavaScript</li>
</ul>
```

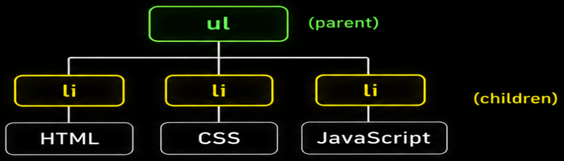
output →



- HTML
- CSS
- JavaScript

DOM Relationship:

** is the parent container && each <li> is a direct child.**



DEFAULT STYLING

Browser injects the following on each <li> by default:

- padding-left: 40px
- list-style-type: disc
- display: list-item

VALID CHILDREN

! Only `<li>` elements are valid direct children of `<ul>`.
Other elements inside are a content model violation.

HTML Attributes

WHAT ARE ATTRIBUTES?

Attributes are **key-value pairs** written inside the **opening tag** that configure or provide additional information about an element.

Tags define *what* an element is — attributes define *how* it behaves, looks, or is identified.

ANATOMY OF AN ATTRIBUTE

```
<h1 class="heading">Web Development</h1>
```

ELEMENT TAG	ATTR NAME	ATTR VALUE	CONTENT
`<h1> … </h1>`	class	"heading"	Web Development

SYNTAX

```
<tag attribute_name="value">Content</tag>
```

- ✓ Attribute is written within the opening tag.
- ✓ Multiple attributes can be used in one tag.
- ✓ Always use quotes around attribute values.
- ✓ Attribute names are case-insensitive in HTML5.

COMMONLY USED ATTRIBUTES (Examples)

`id="main"` `class="box"` `style="color:red;"` `title="Tooltip text"` `href="https://..."` `src="image.png"` and more...

CSS list-style-type

WHAT IT CONTROLS

The **list-style-type** CSS property controls what marker is rendered before each list item.
It can be applied via a CSS file, a `<style>` block, or directly as an inline `style=""` attribute on the `<ul>`.

EXAMPLES **RENDERED RESULT**

```
<ul style="list-style-type: disc;">
   <li>Apple</li>
   <li>Banana</li>
   <li>Cherry</li>
</ul>
```

→

- Apple
- Banana
- Cherry

```
<ul style="list-style-type: square;">
   <li>Apple</li>
   <li>Banana</li>
   <li>Cherry</li>
</ul>
```

→

▪ Apple
▪ Banana
▪ Cherry

```
<ul style="list-style-type: circle;">
   <li>Apple</li>
   <li>Banana</li>
   <li>Cherry</li>
</ul>
```

→

○ Apple
○ Banana
○ Cherry

```
<ul style="list-style-type: none;">
   <li>Apple</li>
   <li>Banana</li>
   <li>Cherry</li>
</ul>
```

→

Apple
Banana
Cherry

Ordered Lists

Use `<ol>` when you have a collection of items where **order does matter** semantically.
The browser renders numbered markers automatically. Swapping items **changes** meaning.

e.g:

```
<ol>
  <li>Boil Water</li>
  <li>Add Pasta</li>
  <li>Drain & Serve</li>
</ol>
```

Output →

BROWSER OUTPUT

1. Boil Water
2. Add Pasta
3. Drain & Serve

Ordered List type="" Attribute

NUMBERING SYSTEMS

The **type** attribute on <ol> changes the **display format** of the counter — but the internal counter is always numeric. The browser converts the number to the requested symbol system for rendering.

```
<ol type="1">  →  → 1. 2. 3. ...
<ol type="A">  →  → A. B. C. ...
<ol type="a">  →  → a. b. c. ...
<ol type="I">  →  → I. II. III. IV. ...
<ol type="i">  →  → i. ii. iii. iv. ...
```

Ordered List start="" Attribute

CORE IDEA

The **start** attribute sets the **initial value of the list counter**. It always accepts a **number**, even when the **type** attribute makes the list display letters or Roman numerals. The browser internally converts the number to the appropriate symbol. **start** always takes an **integer** — never a letter.

```
/* Triathlon stages — starting from Stage 5 */
<ol start="5">
  <li>Swim</li>
  <li>Cycle</li>
  <li>Run</li>
</ol>
```

RENDERS AS (5, 6, 7 ...)

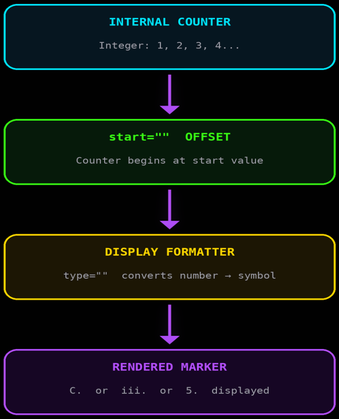
5. Swim
6. Cycle
7. Run

```
/* Exam sections — type="A", start="3" → starts at C */
<ol type="A" start="3">
  <li>Reading</li>
  <li>Writing</li>
  <li>Maths</li>
</ol>
```

RENDERS AS (C, D, E ...)

C. Reading
D. Writing
E. Maths

INTERNAL COUNTER PIPELINE



INTERNAL COUNTER

Integer: 1, 2, 3, 4...

start="" OFFSET

Counter begins at start value

DISPLAY FORMATTER

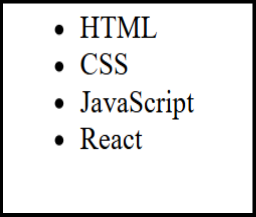
type="" converts number → symbol

RENDERED MARKER

C. or iii. or 5. displayed

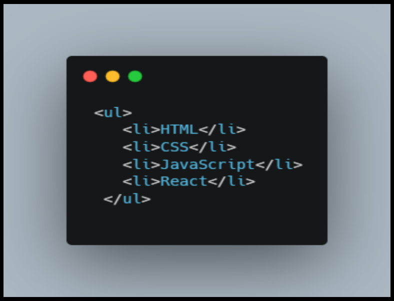
Now it's time to implement these concepts practically:

Practice Tast 1



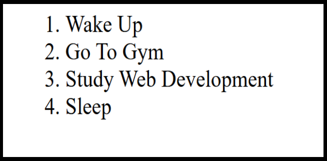
- HTML
- CSS
- JavaScript
- React

Code →



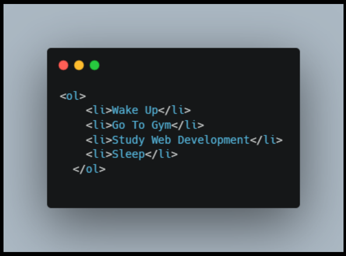
```
<ul>
   <li>HTML</li>
   <li>CSS</li>
   <li>JavaScript</li>
   <li>React</li>
 </ul>
```

Practice Tast 2



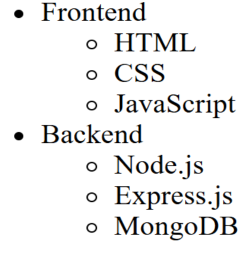
1. Wake Up
2. Go To Gym
3. Study Web Development
4. Sleep

Code →



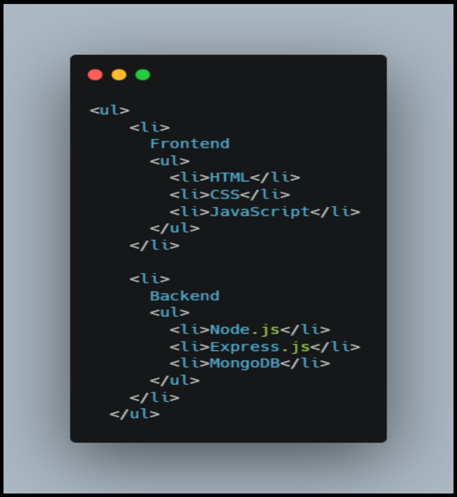
```
<ol>
    <li>Wake Up</li>
    <li>Go To Gym</li>
    <li>Study Web Development</li>
    <li>Sleep</li>
  </ol>
```

Practice Tast 3



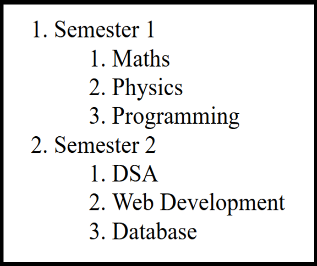
- Frontend
    - HTML
    - CSS
    - JavaScript
- Backend
    - Node.js
    - Express.js
    - MongoDB

Code →



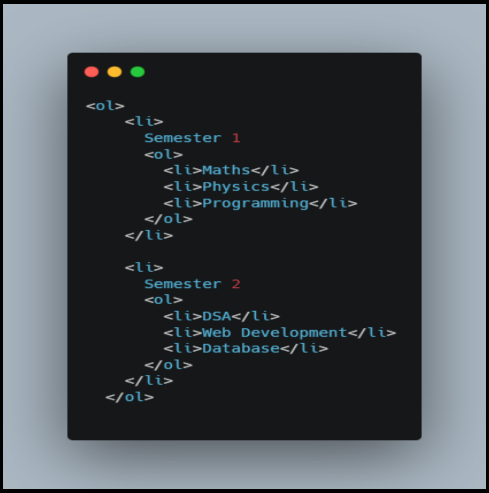
```
<ul>
    <li>
      Frontend
      <ul>
        <li>HTML</li>
        <li>CSS</li>
        <li>JavaScript</li>
      </ul>
    </li>

    <li>
      Backend
      <ul>
        <li>Node.js</li>
        <li>Express.js</li>
        <li>MongoDB</li>
      </ul>
    </li>
  </ul>
```

Practice Tast 4



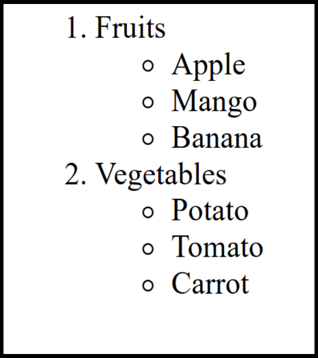
1. Semester 1
    1. Maths
    2. Physics
    3. Programming
2. Semester 2
    1. DSA
    2. Web Development
    3. Database

Code →



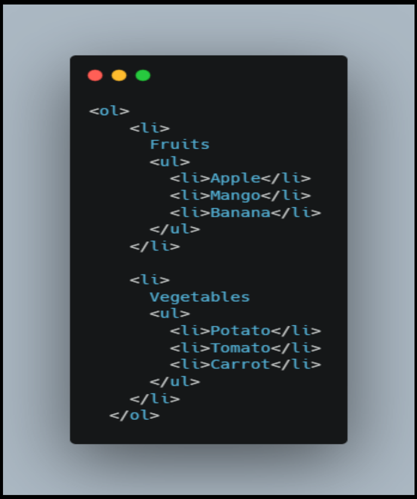
```
<ol>
    <li>
      Semester 1
      <ol>
        <li>Maths</li>
        <li>Physics</li>
        <li>Programming</li>
      </ol>
    </li>

    <li>
      Semester 2
      <ol>
        <li>DSA</li>
        <li>Web Development</li>
        <li>Database</li>
      </ol>
    </li>
  </ol>
```

Practice Tast 5



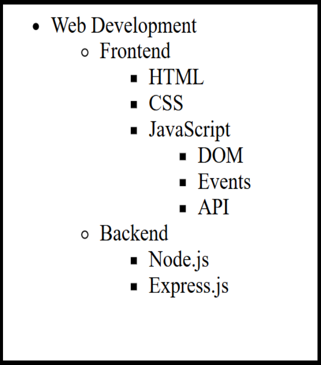
1. Fruits
    - Apple
    - Mango
    - Banana
2. Vegetables
    - Potato
    - Tomato
    - Carrot

Code →



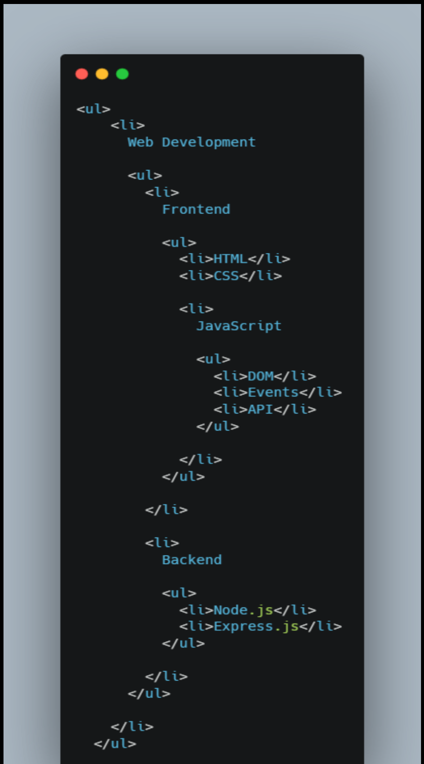
```
<ol>
    <li>
      Fruits
      <ul>
        <li>Apple</li>
        <li>Mango</li>
        <li>Banana</li>
      </ul>
    </li>

    <li>
      Vegetables
      <ul>
        <li>Potato</li>
        <li>Tomato</li>
        <li>Carrot</li>
      </ul>
    </li>
  </ol>
```

Practice Tast 6



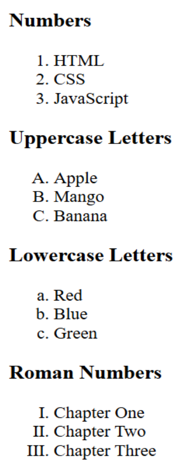
- Web Development
    - Frontend
        - HTML
        - CSS
        - JavaScript
            - DOM
            - Events
            - API
    - Backend
        - Node.js
        - Express.js

Code →



```
<ul>
    <li>
      Web Development

      <ul>
        <li>
          Frontend

          <ul>
            <li>HTML</li>
            <li>CSS</li>

            <li>
              JavaScript

              <ul>
                <li>DOM</li>
                <li>Events</li>
                <li>API</li>
              </ul>

            </li>
          </ul>

        </li>

        <li>
          Backend

          <ul>
            <li>Node.js</li>
            <li>Express.js</li>
          </ul>

        </li>
      </ul>

    </li>
  </ul>
```

Practice Tast 7



Numbers

1. HTML
2. CSS
3. JavaScript

Uppercase Letters

A. Apple
B. Mango
C. Banana

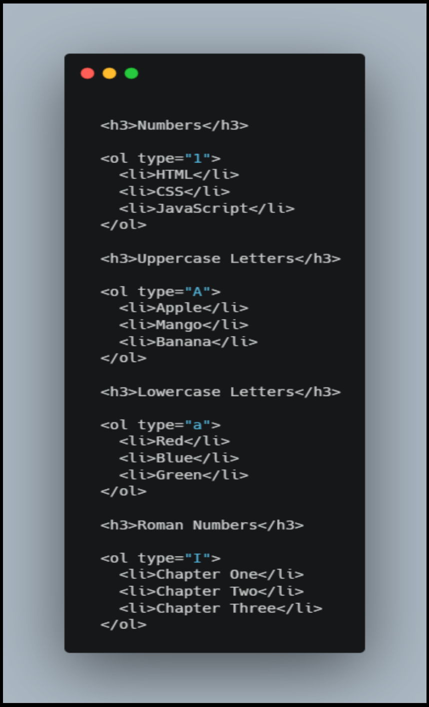
Lowercase Letters

a. Red
b. Blue
c. Green

Roman Numbers

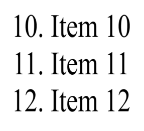
I. Chapter One
II. Chapter Two
III. Chapter Three

Code →



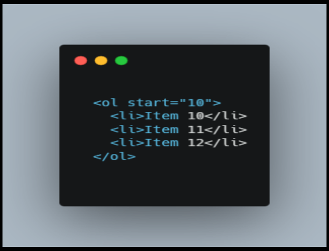
```
<h3>Numbers</h3>

<ol type="1">
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ol>

<h3>Uppercase Letters</h3>

<ol type="A">
  <li>Apple</li>
  <li>Mango</li>
  <li>Banana</li>
</ol>

<h3>Lowercase Letters</h3>

<ol type="a">
  <li>Red</li>
  <li>Blue</li>
  <li>Green</li>
</ol>

<h3>Roman Numbers</h3>

<ol type="I">
  <li>Chapter One</li>
  <li>Chapter Two</li>
  <li>Chapter Three</li>
</ol>
```

Practice Tast 8



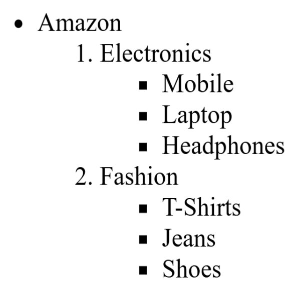
10. Item 10
11. Item 11
12. Item 12

Code →



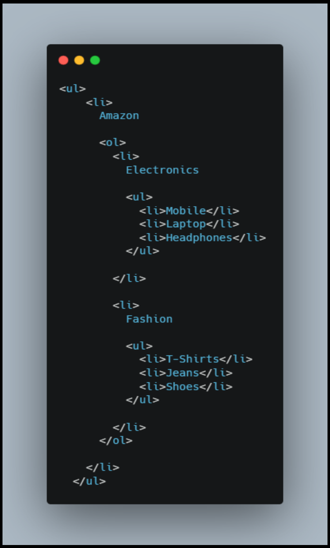
```
<ol start="10">
  <li>Item 10</li>
  <li>Item 11</li>
  <li>Item 12</li>
</ol>
```

Practice Tast 9



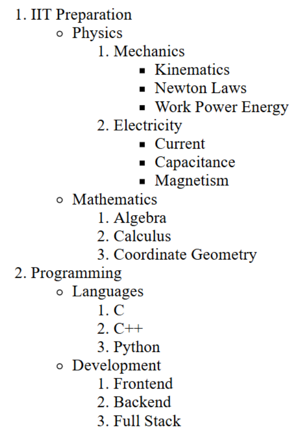
- Amazon
  1. Electronics
     - Mobile
     - Laptop
     - Headphones
  2. Fashion
     - T-Shirts
     - Jeans
     - Shoes

Code →



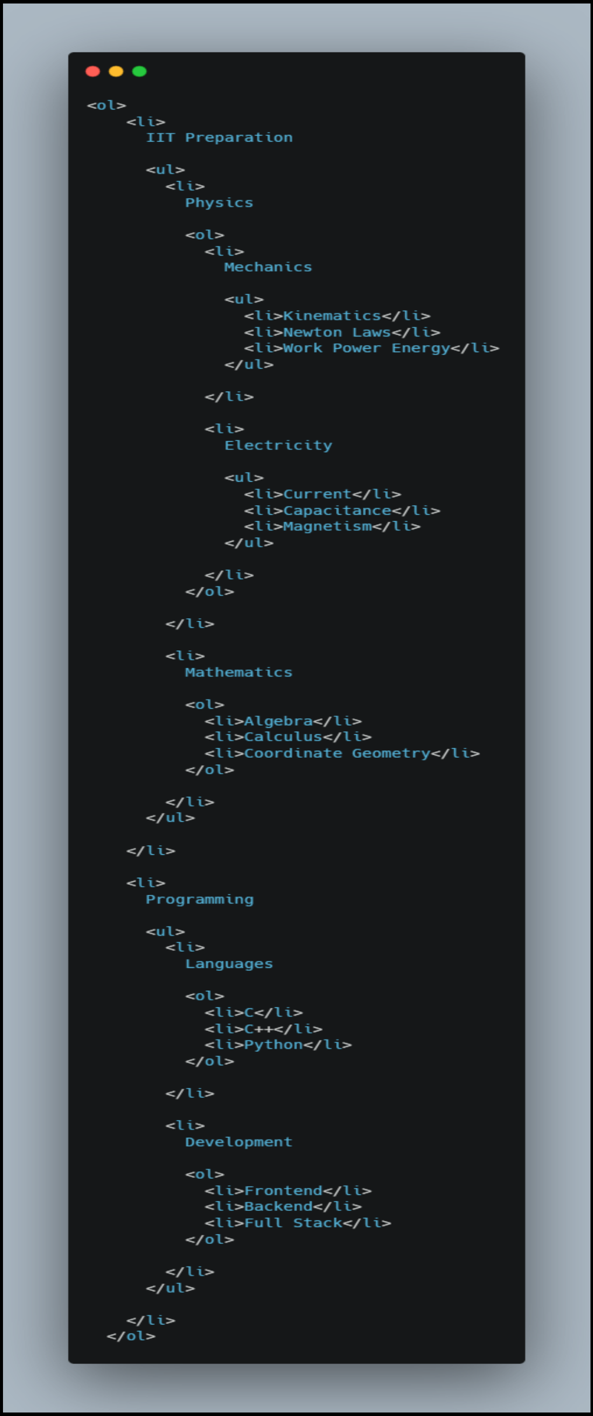
```
<ul>
    <li>
      Amazon

      <ol>
        <li>
          Electronics

          <ul>
            <li>Mobile</li>
            <li>Laptop</li>
            <li>Headphones</li>
          </ul>

        </li>

        <li>
          Fashion

          <ul>
            <li>T-Shirts</li>
            <li>Jeans</li>
            <li>Shoes</li>
          </ul>

        </li>
      </ol>

    </li>
  </ul>
```

Practice Tast 10



1. IIT Preparation
    - Physics
        1. Mechanics
            - Kinematics
            - Newton Laws
            - Work Power Energy
        2. Electricity
            - Current
            - Capacitance
            - Magnetism
    - Mathematics
        1. Algebra
        2. Calculus
        3. Coordinate Geometry
2. Programming
    - Languages
        1. C
        2. C++
        3. Python
    - Development
        1. Frontend
        2. Backend
        3. Full Stack

Code



```
<ol>
    <li>
      IIT Preparation

      <ul>
        <li>
          Physics

          <ol>
            <li>
              Mechanics

              <ul>
                <li>Kinematics</li>
                <li>Newton Laws</li>
                <li>Work Power Energy</li>
              </ul>

            </li>

            <li>
              Electricity

              <ul>
                <li>Current</li>
                <li>Capacitance</li>
                <li>Magnetism</li>
              </ul>

            </li>
          </ol>

        </li>

        <li>
          Mathematics

          <ol>
            <li>Algebra</li>
            <li>Calculus</li>
            <li>Coordinate Geometry</li>
          </ol>

        </li>
      </ul>

    </li>

    <li>
      Programming

      <ul>
        <li>
          Languages

          <ol>
            <li>C</li>
            <li>C++</li>
            <li>Python</li>
          </ol>

        </li>

        <li>
          Development

          <ol>
            <li>Frontend</li>
            <li>Backend</li>
            <li>Full Stack</li>
          </ol>

        </li>
      </ul>

    </li>
  </ol>
```

Note:

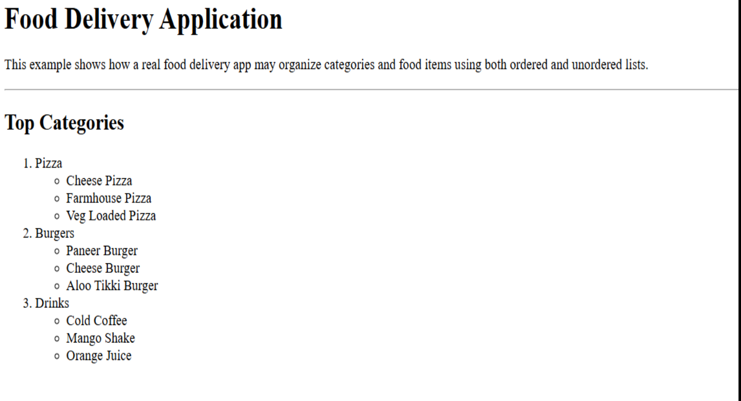
•In an unordered list (<ul>), the browser uses a disc marker (•) by default for each list item (<li>).

•Nested lists are also possible, which means a <ul> or <ol> can be placed inside another <li> element to create multi-level lists.

•Browsers automatically handle the indentation and bullet styling of nested lists, making the content easier to read and visually structured.

•The appearance of list markers can be changed using the CSS list-style-type property.
For example: disc, circle, square, or none.

Practice Task 11



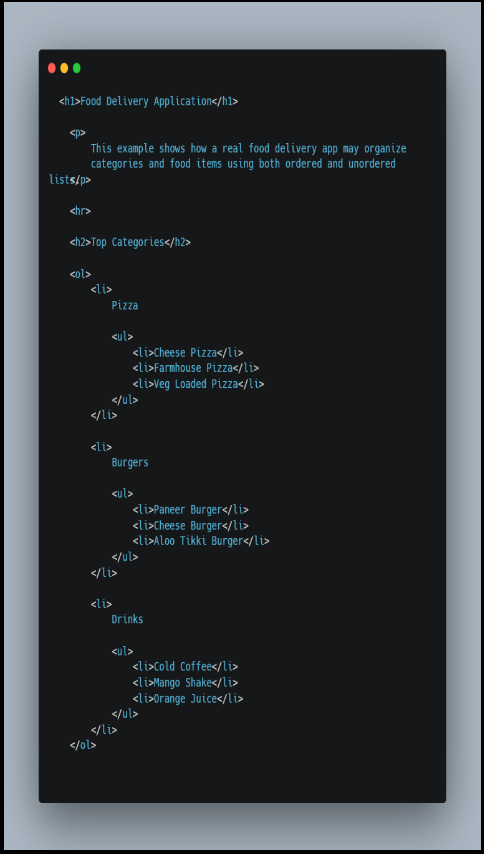
Food Delivery Application

This example shows how a real food delivery app may organize categories and food items using both ordered and unordered lists.

Top Categories

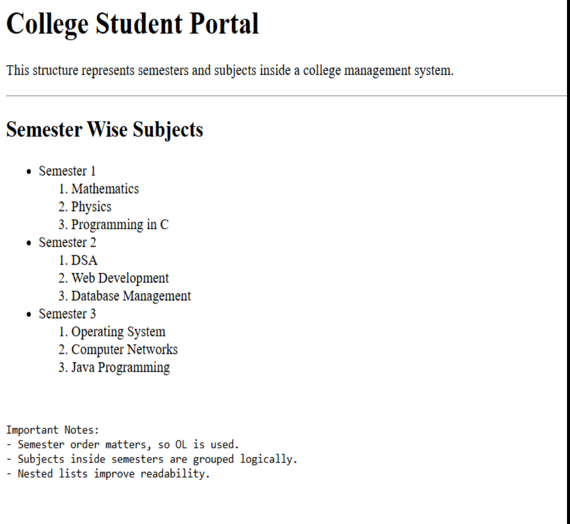
1. Pizza
    - Cheese Pizza
    - Farmhouse Pizza
    - Veg Loaded Pizza
2. Burgers
    - Paneer Burger
    - Cheese Burger
    - Aloo Tikki Burger
3. Drinks
    - Cold Coffee
    - Mango Shake
    - Orange Juice

Code →



```
  <h1>Food Delivery Application</h1>

    <p>
        This example shows how a real food delivery app may organize
        categories and food items using both ordered and unordered
lists</p>

    <hr>

    <h2>Top Categories</h2>

    <ol>
        <li>
            Pizza

            <ul>
                <li>Cheese Pizza</li>
                <li>Farmhouse Pizza</li>
                <li>Veg Loaded Pizza</li>
            </ul>
        </li>

        <li>
            Burgers

            <ul>
                <li>Paneer Burger</li>
                <li>Cheese Burger</li>
                <li>Aloo Tikki Burger</li>
            </ul>
        </li>

        <li>
            Drinks

            <ul>
                <li>Cold Coffee</li>
                <li>Mango Shake</li>
                <li>Orange Juice</li>
            </ul>
        </li>
    </ol>
```

Practice Tast 12



College Student Portal

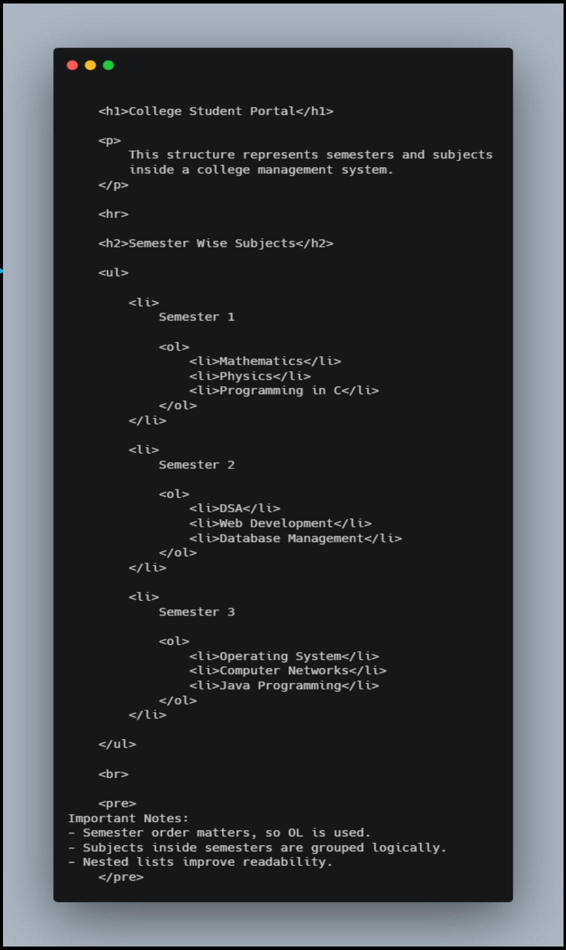
This structure represents semesters and subjects inside a college management system.

Semester Wise Subjects

- Semester 1
    1. Mathematics
    2. Physics
    3. Programming in C
- Semester 2
    1. DSA
    2. Web Development
    3. Database Management
- Semester 3
    1. Operating System
    2. Computer Networks
    3. Java Programming

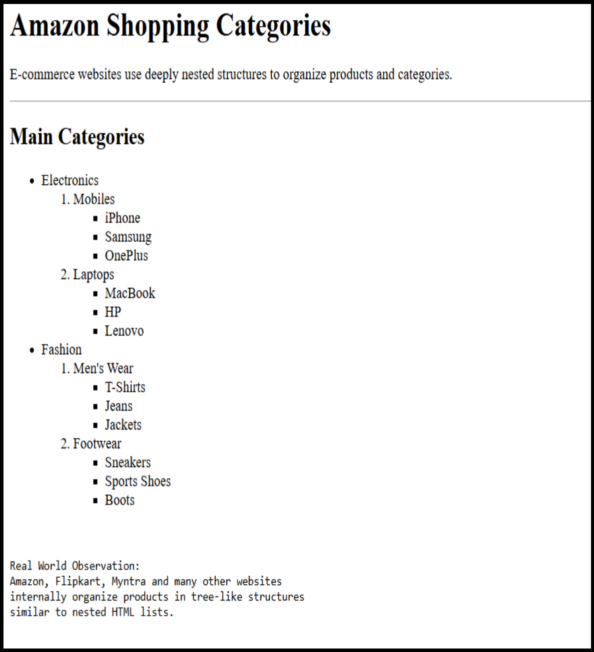
```
Important Notes:
- Semester order matters, so OL is used.
- Subjects inside semesters are grouped logically.
- Nested lists improve readability.
```

Code →



```
    <h1>College Student Portal</h1>

    <p>
        This structure represents semesters and subjects
        inside a college management system.
    </p>

    <hr>

    <h2>Semester Wise Subjects</h2>

    <ul>

        <li>
            Semester 1

            <ol>
                <li>Mathematics</li>
                <li>Physics</li>
                <li>Programming in C</li>
            </ol>
        </li>

        <li>
            Semester 2

            <ol>
                <li>DSA</li>
                <li>Web Development</li>
                <li>Database Management</li>
            </ol>
        </li>

        <li>
            Semester 3

            <ol>
                <li>Operating System</li>
                <li>Computer Networks</li>
                <li>Java Programming</li>
            </ol>
        </li>

    </ul>

    <br>

    <pre>
Important Notes:
- Semester order matters, so OL is used.
- Subjects inside semesters are grouped logically.
- Nested lists improve readability.
    </pre>
```

Practice Task 13



Amazon Shopping Categories

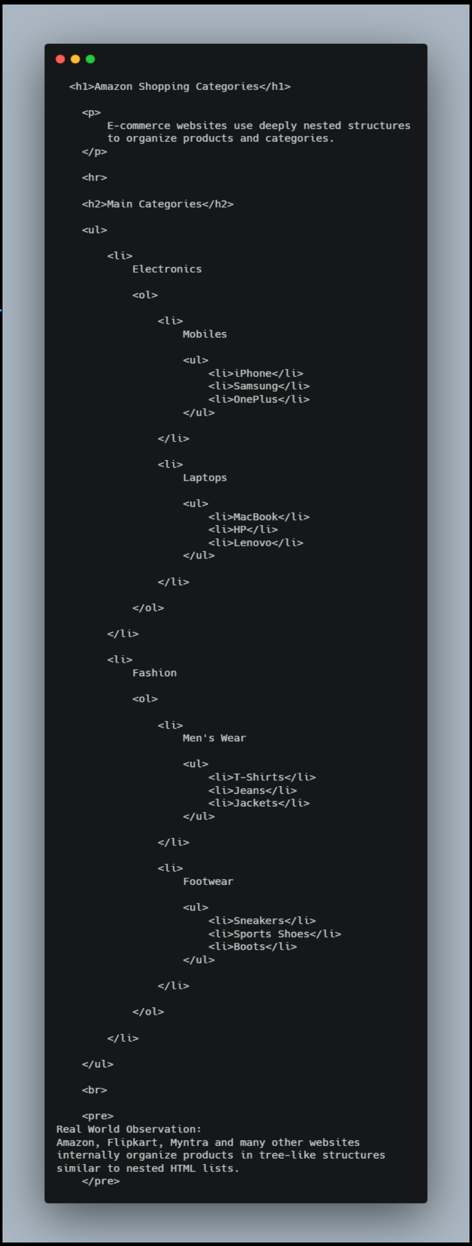
E-commerce websites use deeply nested structures to organize products and categories.

Main Categories

- Electronics
    1. Mobiles
        - iPhone
        - Samsung
        - OnePlus
    2. Laptops
        - MacBook
        - HP
        - Lenovo
- Fashion
    1. Men's Wear
        - T-Shirts
        - Jeans
        - Jackets
    2. Footwear
        - Sneakers
        - Sports Shoes
        - Boots

```
Real World Observation:
Amazon, Flipkart, Myntra and many other websites
internally organize products in tree-like structures
similar to nested HTML lists.
```

Code



```
  <h1>Amazon Shopping Categories</h1>

    <p>
        E-commerce websites use deeply nested structures
        to organize products and categories.
    </p>

    <hr>

    <h2>Main Categories</h2>

    <ul>

        <li>
            Electronics

            <ol>

                <li>
                    Mobiles

                    <ul>
                        <li>iPhone</li>
                        <li>Samsung</li>
                        <li>OnePlus</li>
                    </ul>

                </li>

                <li>
                    Laptops

                    <ul>
                        <li>MacBook</li>
                        <li>HP</li>
                        <li>Lenovo</li>
                    </ul>

                </li>

            </ol>

        </li>

        <li>
            Fashion

            <ol>

                <li>
                    Men's Wear

                    <ul>
                        <li>T-Shirts</li>
                        <li>Jeans</li>
                        <li>Jackets</li>
                    </ul>

                </li>

                <li>
                    Footwear

                    <ul>
                        <li>Sneakers</li>
                        <li>Sports Shoes</li>
                        <li>Boots</li>
                    </ul>

                </li>

            </ol>

        </li>

    </ul>

    <br>

    <pre>
Real World Observation:
Amazon, Flipkart, Myntra and many other websites
internally organize products in tree-like structures
similar to nested HTML lists.
    </pre>
```